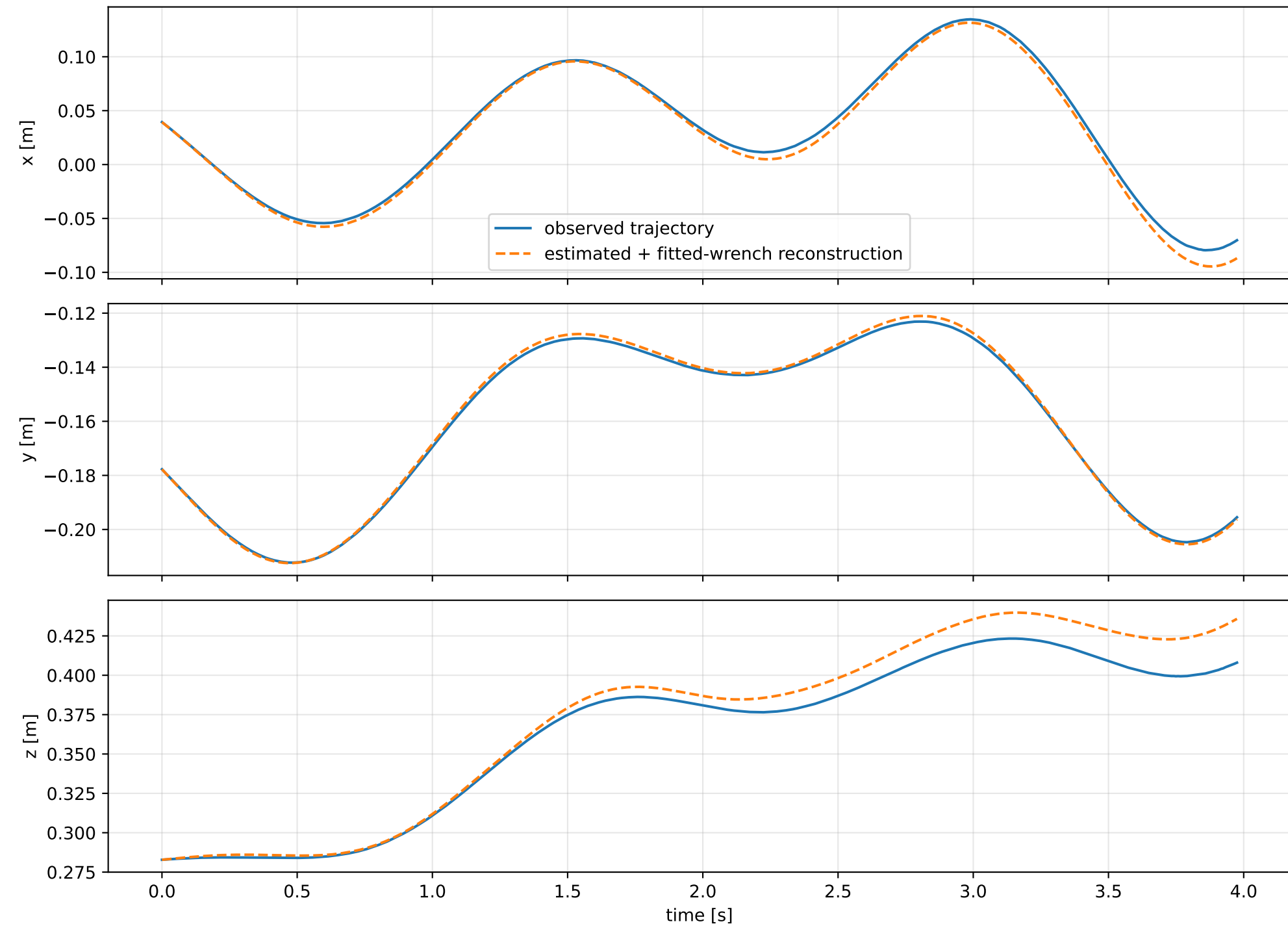
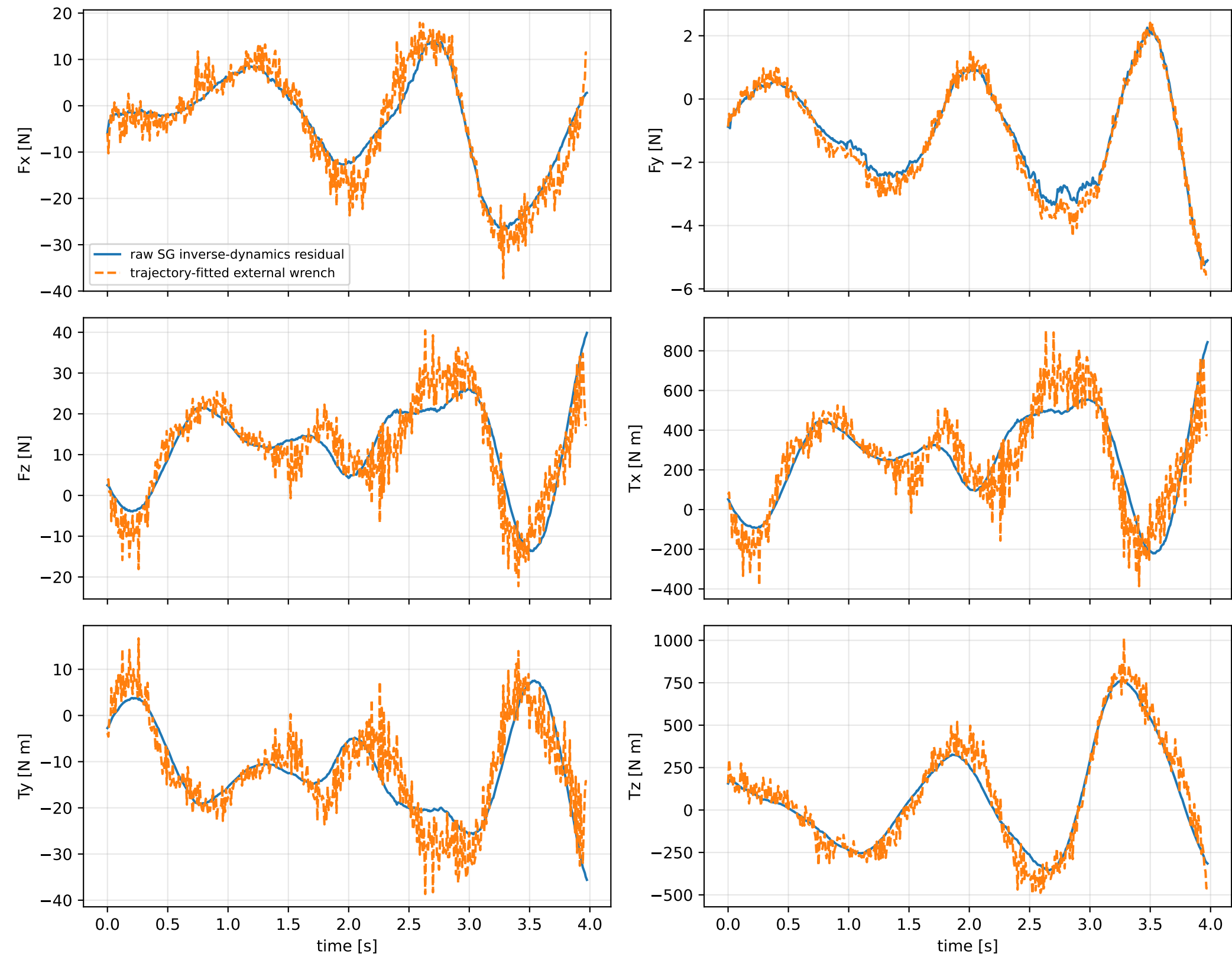


sweep_sg_window_covariance_004: trajectory

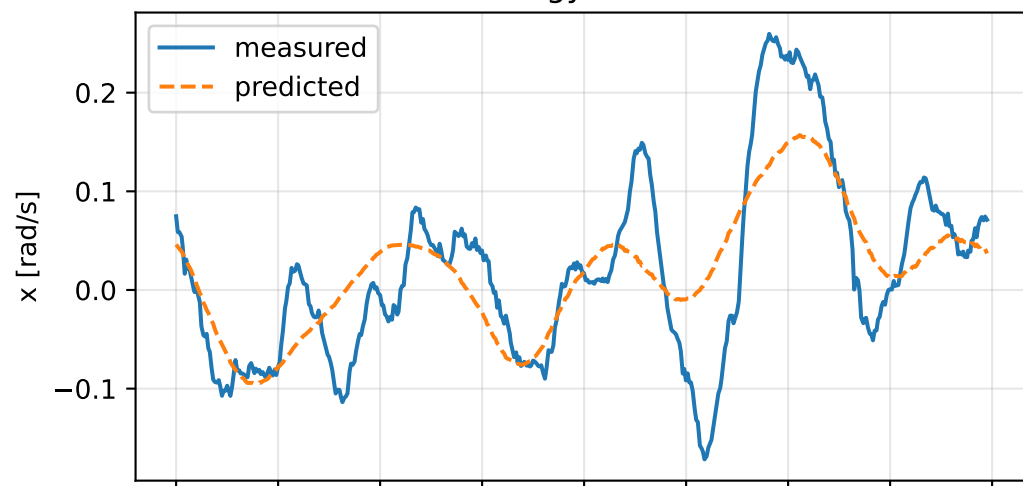


sweep_sg_window_covariance_004: residual wrench

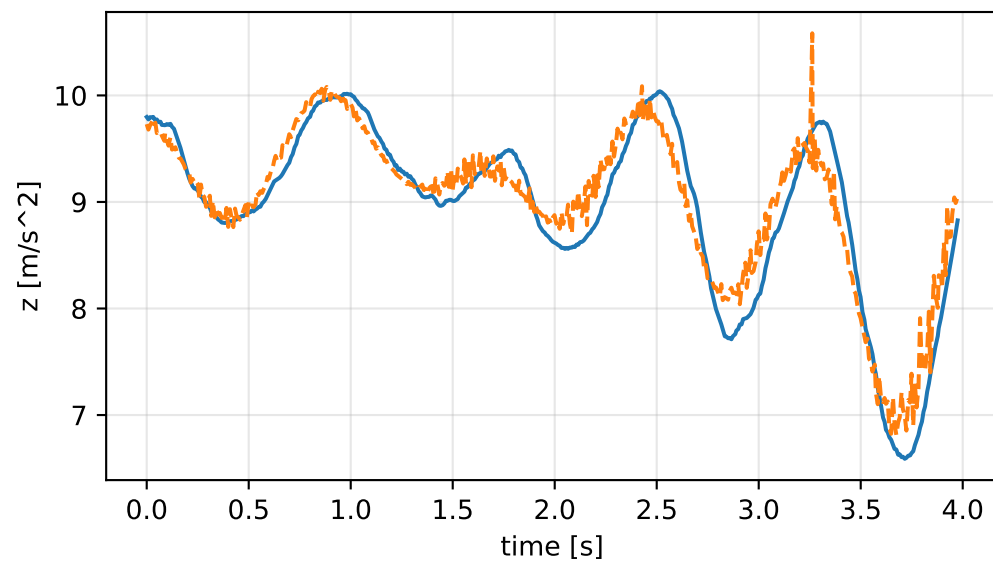
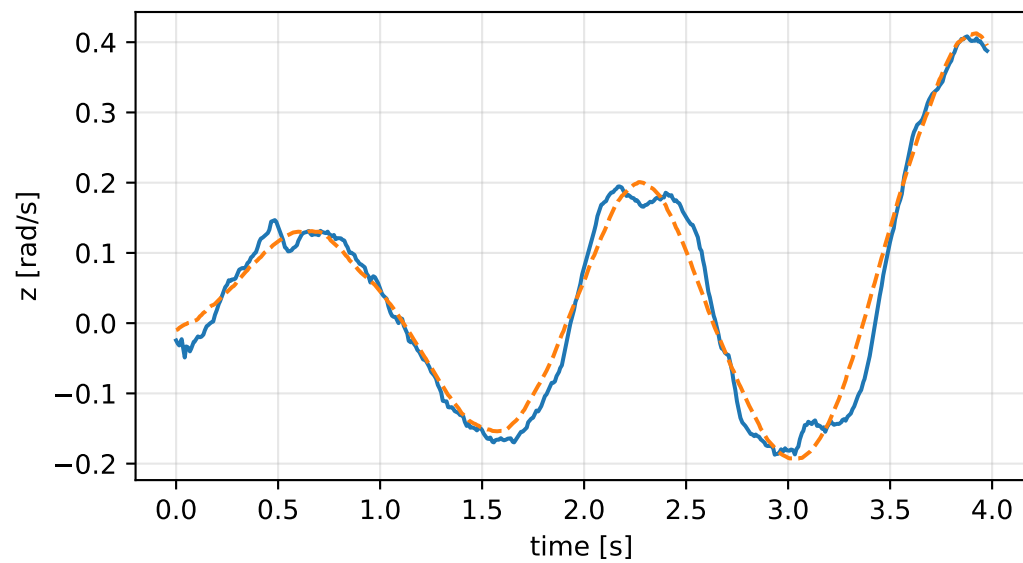
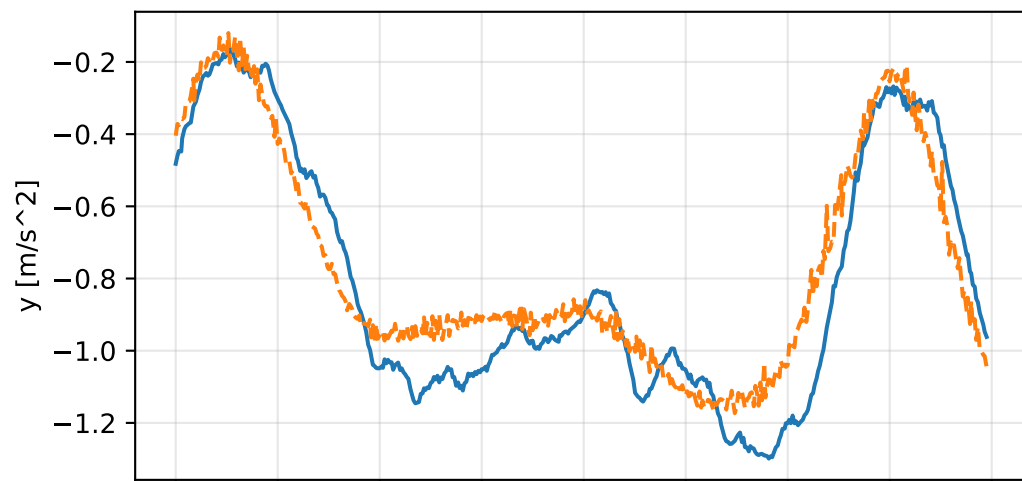
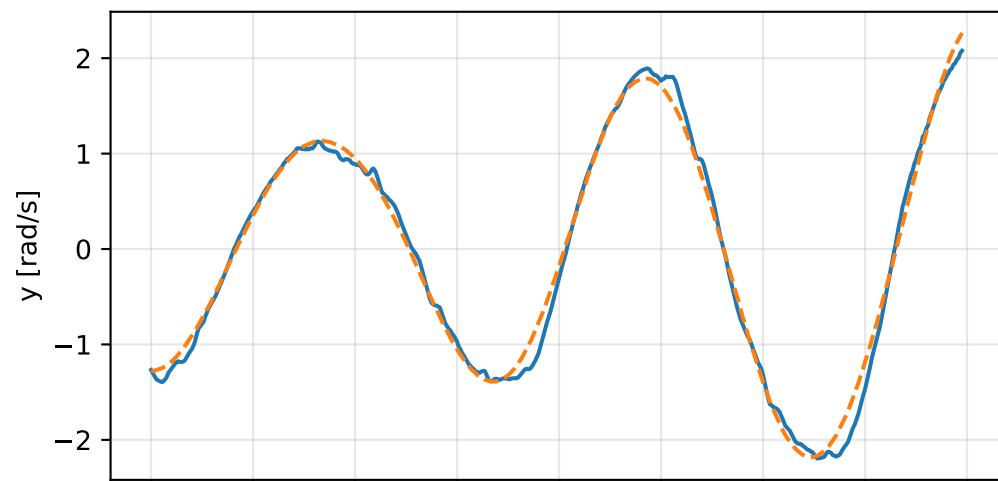
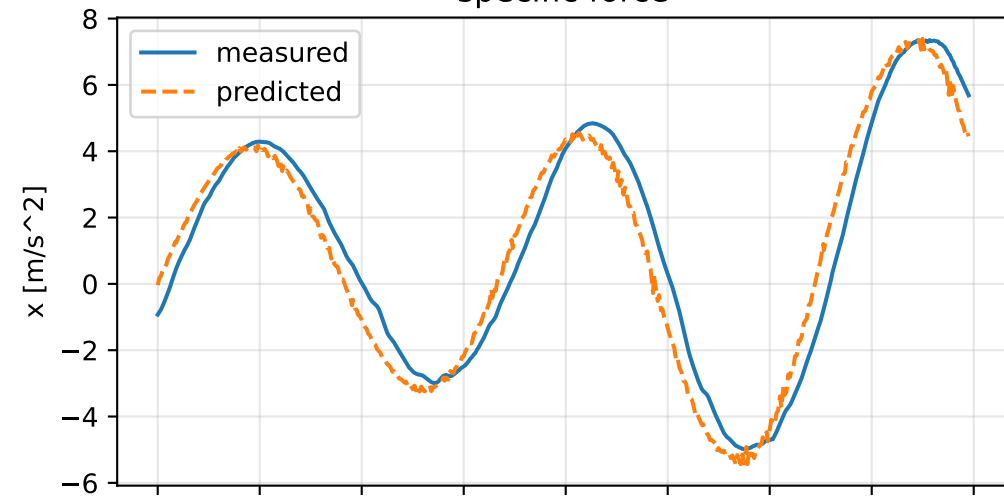


sweep_sg_window_covariance_004: measured sensor comparison

gyro



specific force



sweep_sg_window_covariance__004: nominal -> estimated parameters

Case: sweep_sg_window_covariance__004
status: completed (all stages completed)

mass [kg] nominal 2.35159759 estimated 4.40116524 delta 2.04956765

inertia [kg m^2] (nominal -> estimated; delta)

[6.5000061e-02 -7.2789925e-07 1.9015080e-05] -> [6.4409109e+02 -2.8110798e+01 5.5360026e-02] d=[6.4402609e+02 -2.8110798e+01 5.5360026e-02]
[-7.2789925e-07 6.4952656e-02 5.9167305e-08] -> [-28.1107978 1.6295504 -2.9669222] d=[-28.1107971 1.5645978 -2.9669222]
[1.9015080e-05 5.9167305e-08 1.2899211e-01] -> [5.5360026e-02 -2.9669222e+00 6.4542411e+02] d=[5.5341011e-02 -2.9669222e+00 6.4542411e+02]

CoG [m] [-2.0247086e-03 -3.0526579e-05 9.5097496e-03] -> [0.3231315 6.768144 -0.028148] d=[0.3251562 6.7681745 -0.028148]

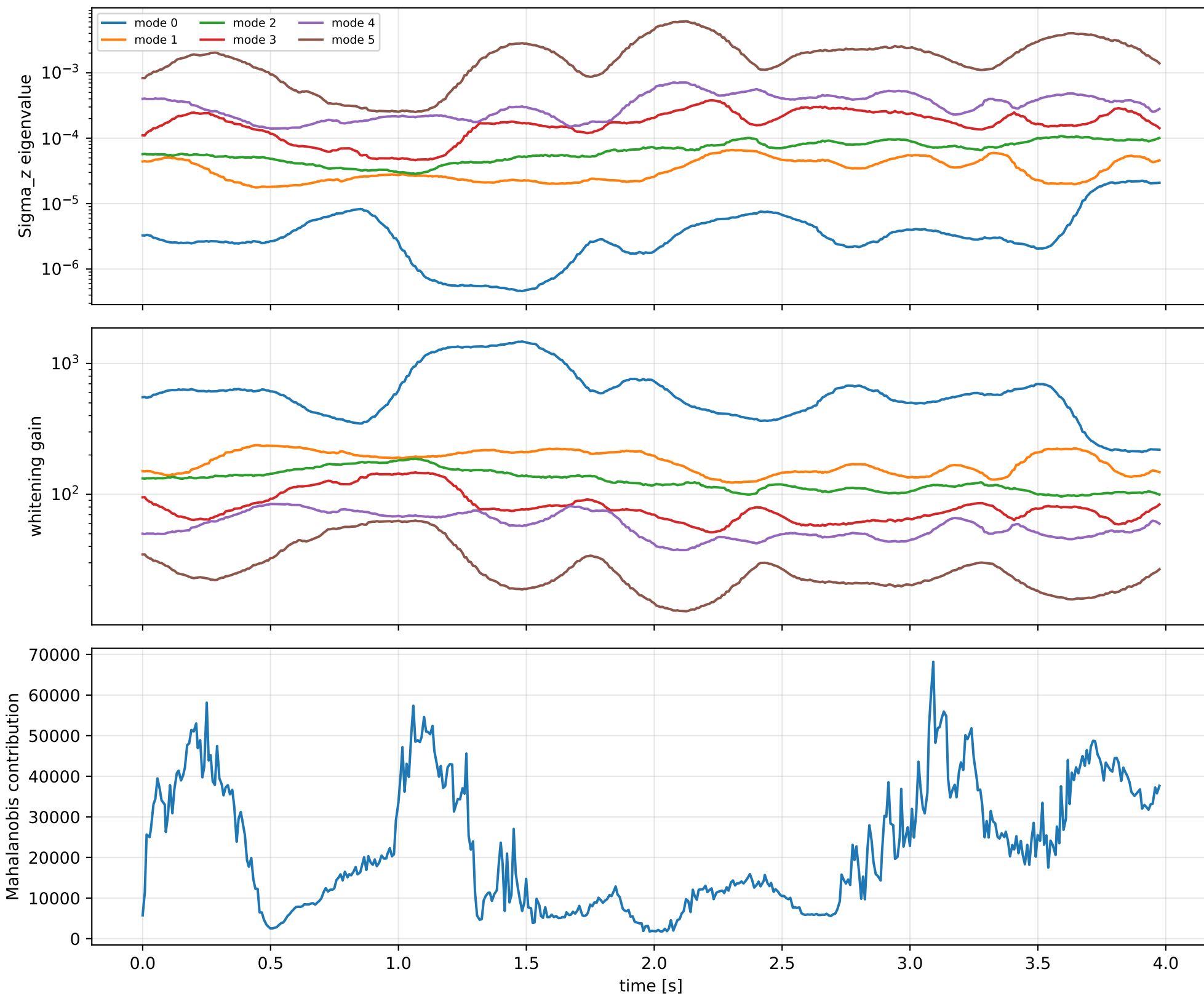
rotor force effectiveness

[1. 1. 1. 1.] -> [2.2169095e+00 2.3619320e+00 4.4225074e-01 2.8332290e-05] d=[1.2169095 1.361932 -0.5577493 -0.9999717]

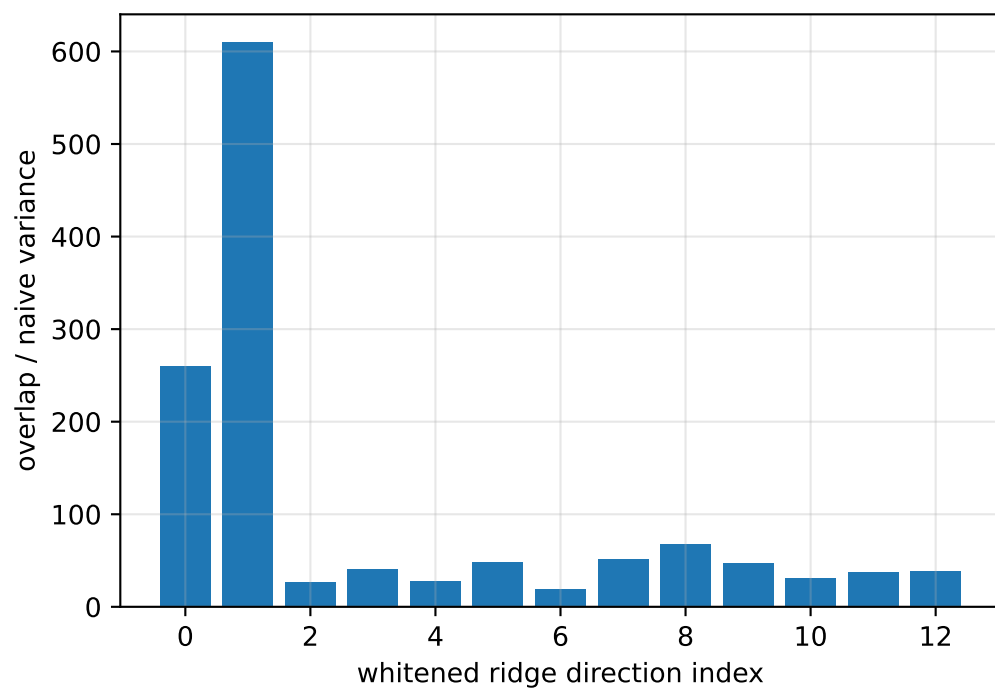
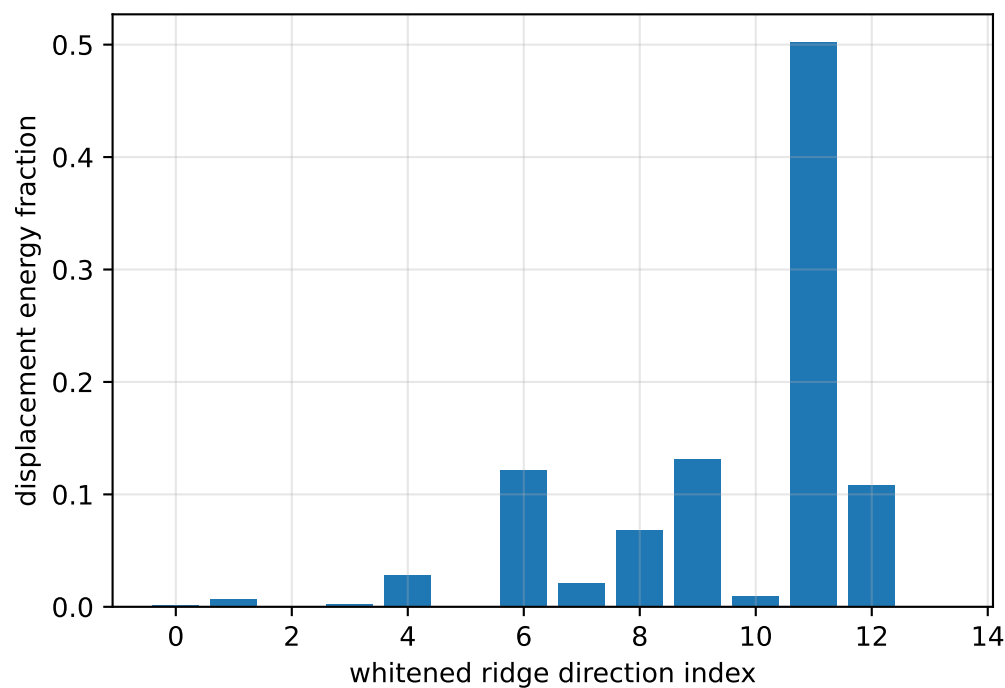
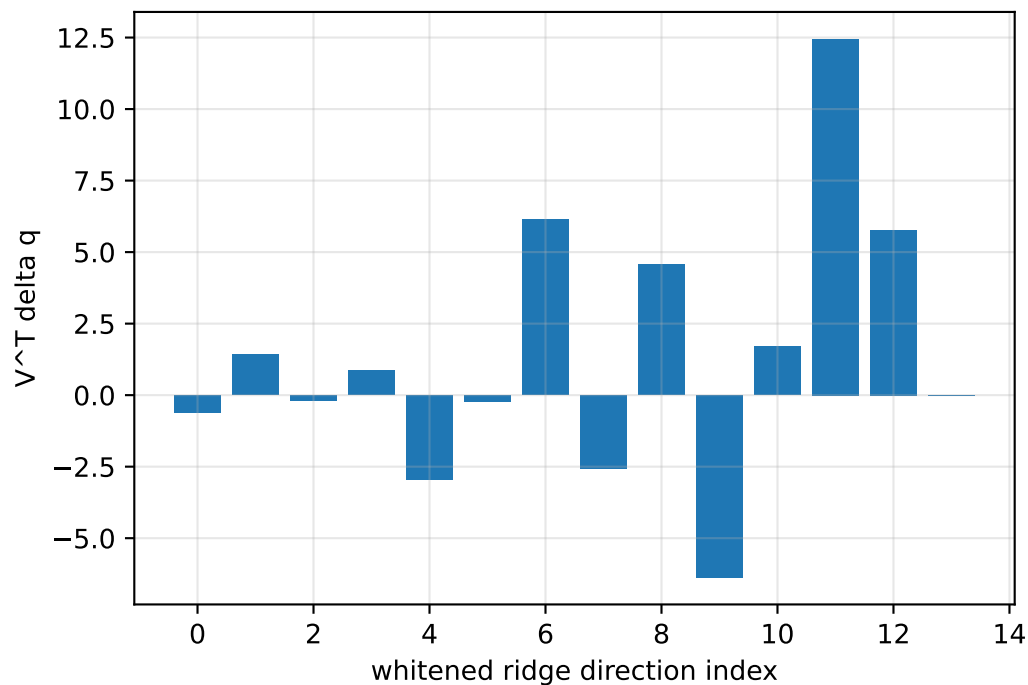
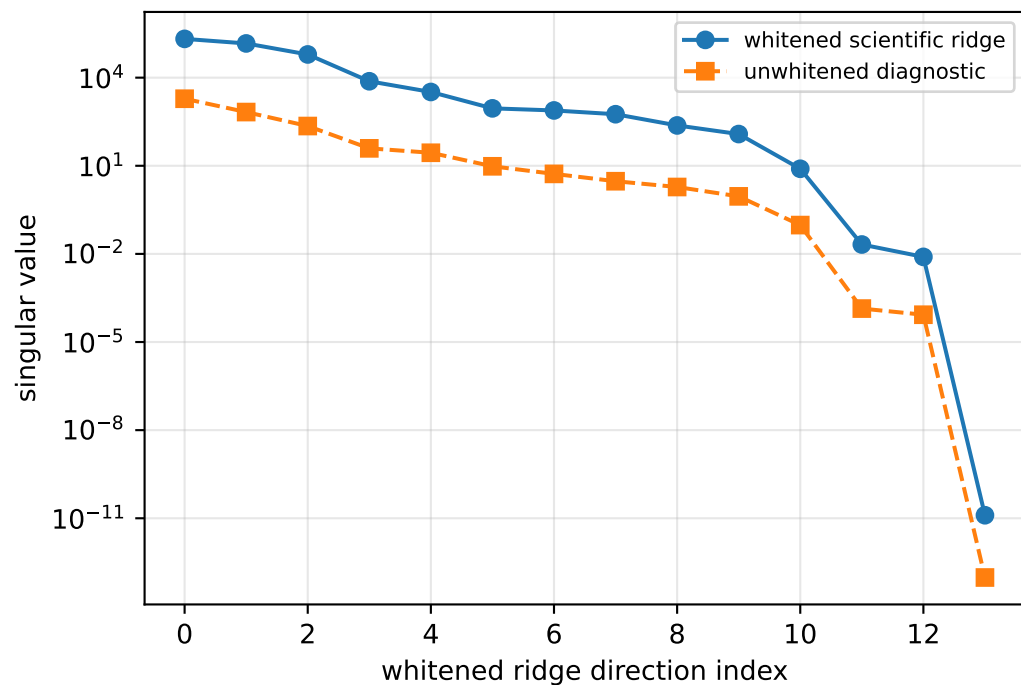
rotor lag [s] 0.2

gimbal lag [s] 0.119968414

sweep_sg_window_covariance_004: covariance weighting

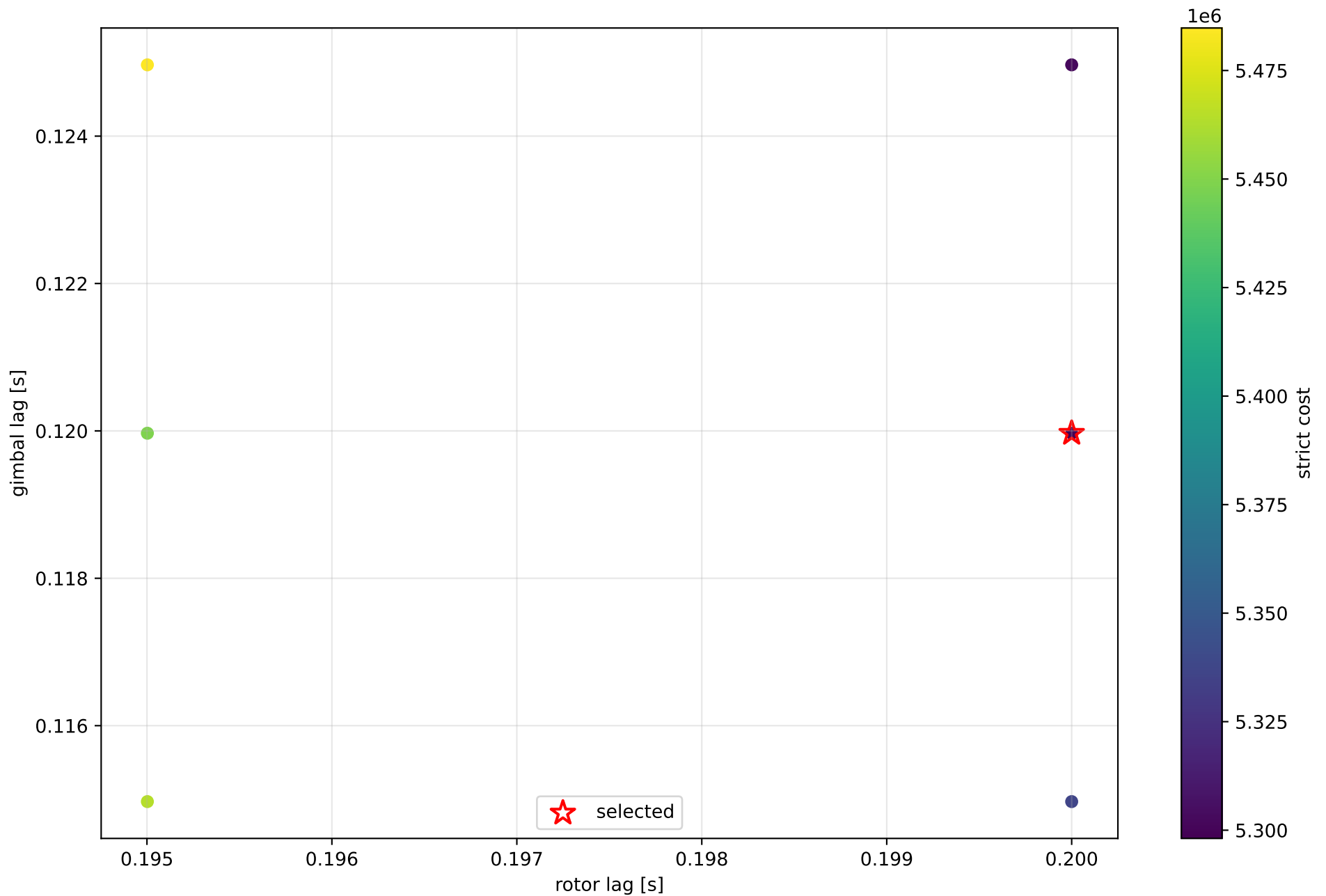


sweep_sg_window_covariance_004: parameter ridge



exact scale gauge: || v_{scale} ||=3.21887e-10, rank=13, nullity=1

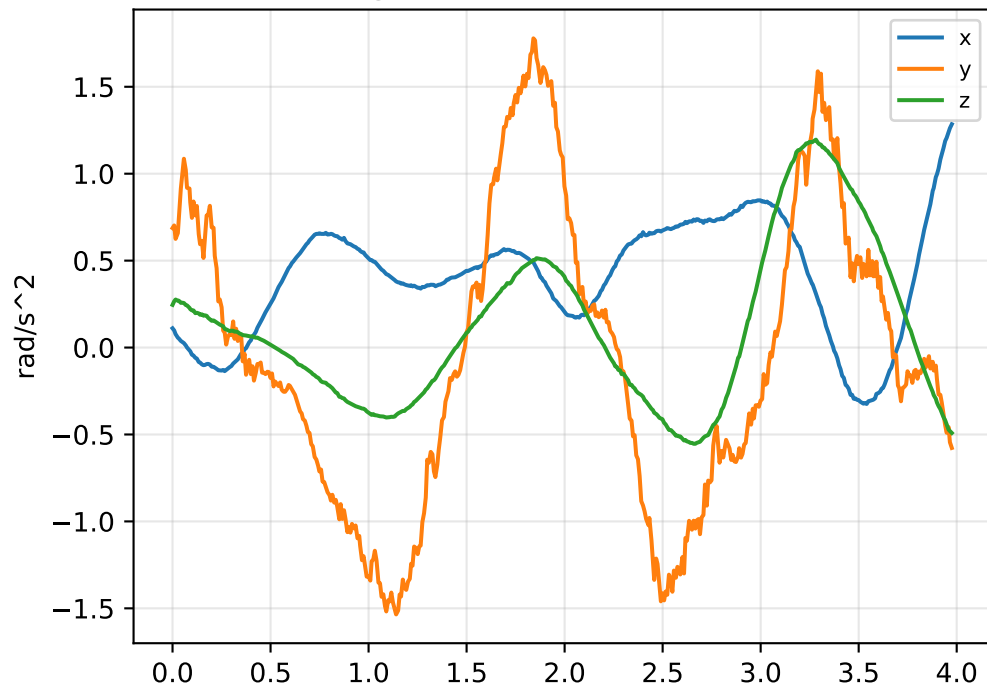
sweep_sg_window_covariance__004: strict lag candidates



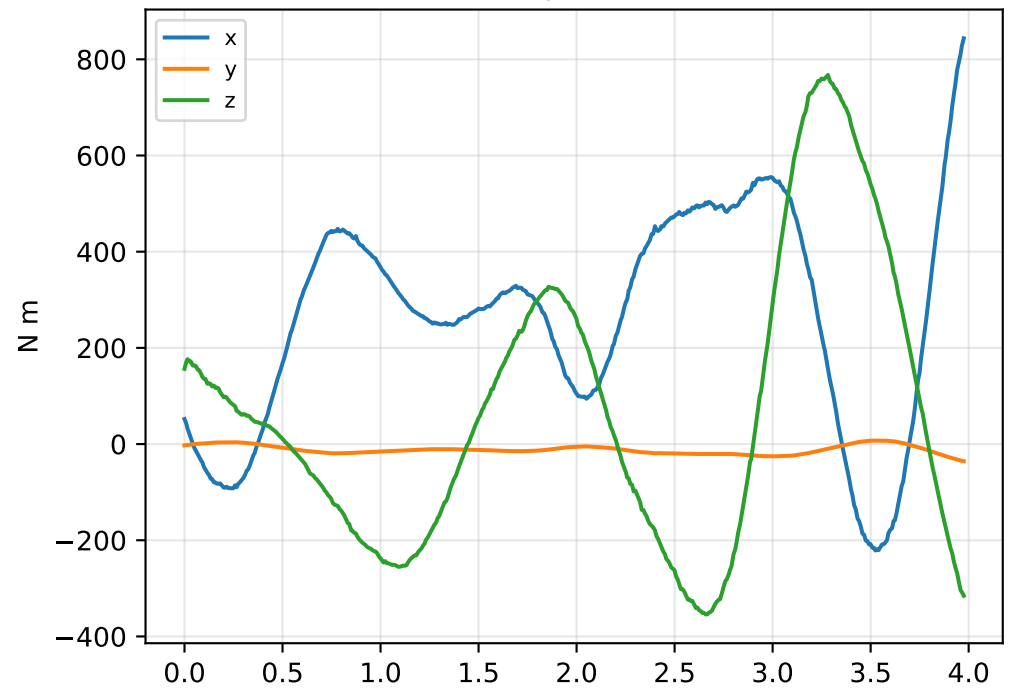
boundary flags: rotor lower=False upper=True; gimbal lower=False upper=False

sweep_sg_window_covariance_004: acceleration/wrench closure

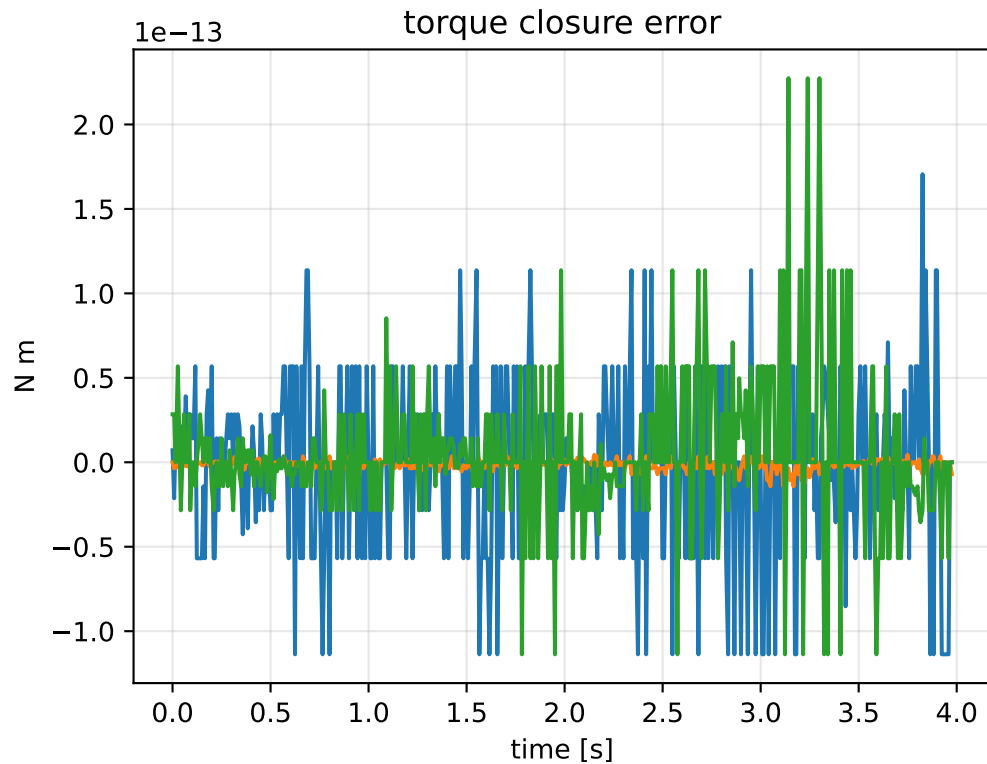
angular-acceleration residual



raw torque residual



torque closure error



principal moments [kg m²]
 nominal: [0.064953 0.065 0.128992]
 estimated: [3.883144e-01 6.451841e+02 6.455723e+02]
 estimated / nominal: [5.978423e+00 9.925899e+03 5.004742e+03]

force closure max/rms: 1.42109e-14 / 3.09359e-15
 torque closure max/rms: 2.27374e-13 / 3.64416e-14